

# Three books, one account

A resident short-premium book, two transient tenants, and a capital boundary that stops them fighting over the same margin. This is the whole system, from the outer frame down to each gate.

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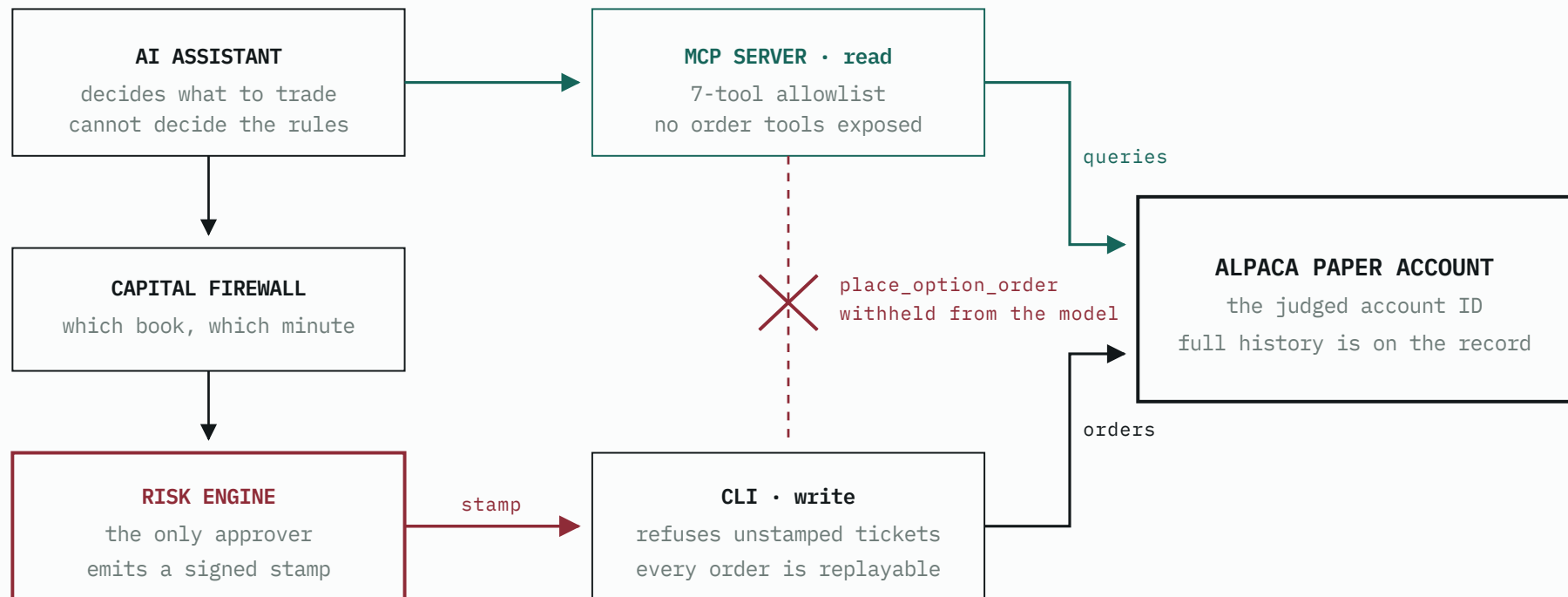
**Repo** `alpaca-hackathon`   **Books** `carry` · `intraday` · `opportunistic`   **Tests** `181 offline`   **Judged window** `28 Aug – 4 Sep 2026`

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## THE OUTER FRAME

### One account is the only shared resource

Everything else in the system exists to decide who may touch it, with how much, and at which minute. Alpaca is reached three ways, deliberately, for three different jobs.

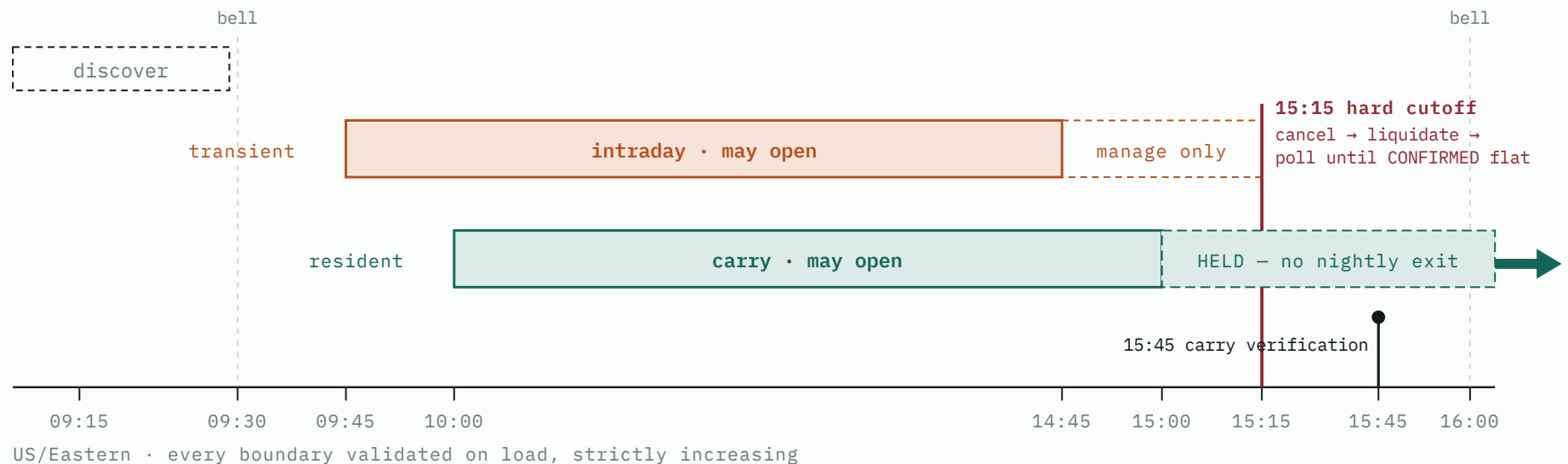


**Reads and writes take different paths on purpose.** The agent queries Alpaca's MCP server directly and reasons over live state — that is real autonomy, not a scripted pipeline with a language model bolted on. But the server's own order-placing tools are never exposed to it. An agent that can place a raw order can bypass the firewall, and the whole point of the firewall is that nothing can.

# One resident, two transient tenants

The carry book and the intraday book are uncorrelated in signal and opposite in vol exposure. That is the real portfolio argument for running both — and the reason the boundary between them has to be enforced rather than assumed.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>RESIDENT</div> <div><b>carry</b></div> <div>Sells the IV–RV spread as defined-risk structures and <b>holds them</b>. Theta accrues on calendar days, weekend included; a nightly round trip would pay the spread for nothing.</div> <div>EDGE</div> <div>variance risk premium</div> <div>STRUCTURE</div> <div>iron condor · credit vertical · calendar</div> <div>HOLD</div> <div>3–10 sessions, 7–14 DTE</div> <div>WINDOW ET</div> <div>10:00 – 15:00</div> <div>RISK</div> <div>short gamma into a move</div> | <div>TRANSIENT</div> <div><b>intraday</b></div> <div>Momentum expressed <i>through</i> options. The option is leverage and defined risk — it is not the source of edge. Flat by 15:15, every session, without exception.</div> <div>EDGE</div> <div>directional continuation</div> <div>STRUCTURE</div> <div>long option · debit vertical</div> <div>HOLD</div> <div>minutes to hours, 0–2 DTE</div> <div>WINDOW ET</div> <div>09:45 – 14:45</div> <div>RISK</div> <div>spread cost, signal decay</div> | <div>TRANSIENT</div> <div><b>opportunistic</b></div> <div>Dormant unless a dated print is due. It may never fire inside the judged week — an agent that correctly stands down is demonstrating judgement, and that is reported as the outcome.</div> <div>EDGE</div> <div>event premium mispricing</div> <div>STRUCTURE</div> <div>iron condor, SPY/QQQ only</div> <div>HOLD</div> <div>1–3 sessions</div> <div>WINDOW ET</div> <div>09:45 – 14:45</div> <div>RISK</div> <div>the print itself</div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



**The transient books go home; the resident book does not.** The 15:15 cutoff liquidates only the legs the position ledger attributes to a transient book, so a multi-session iron condor is never collateral damage of the day book closing. At 15:45 the day is signed off: zero transient exposure, zero working orders, a fresh Reg T read, carry margin covered with headroom.

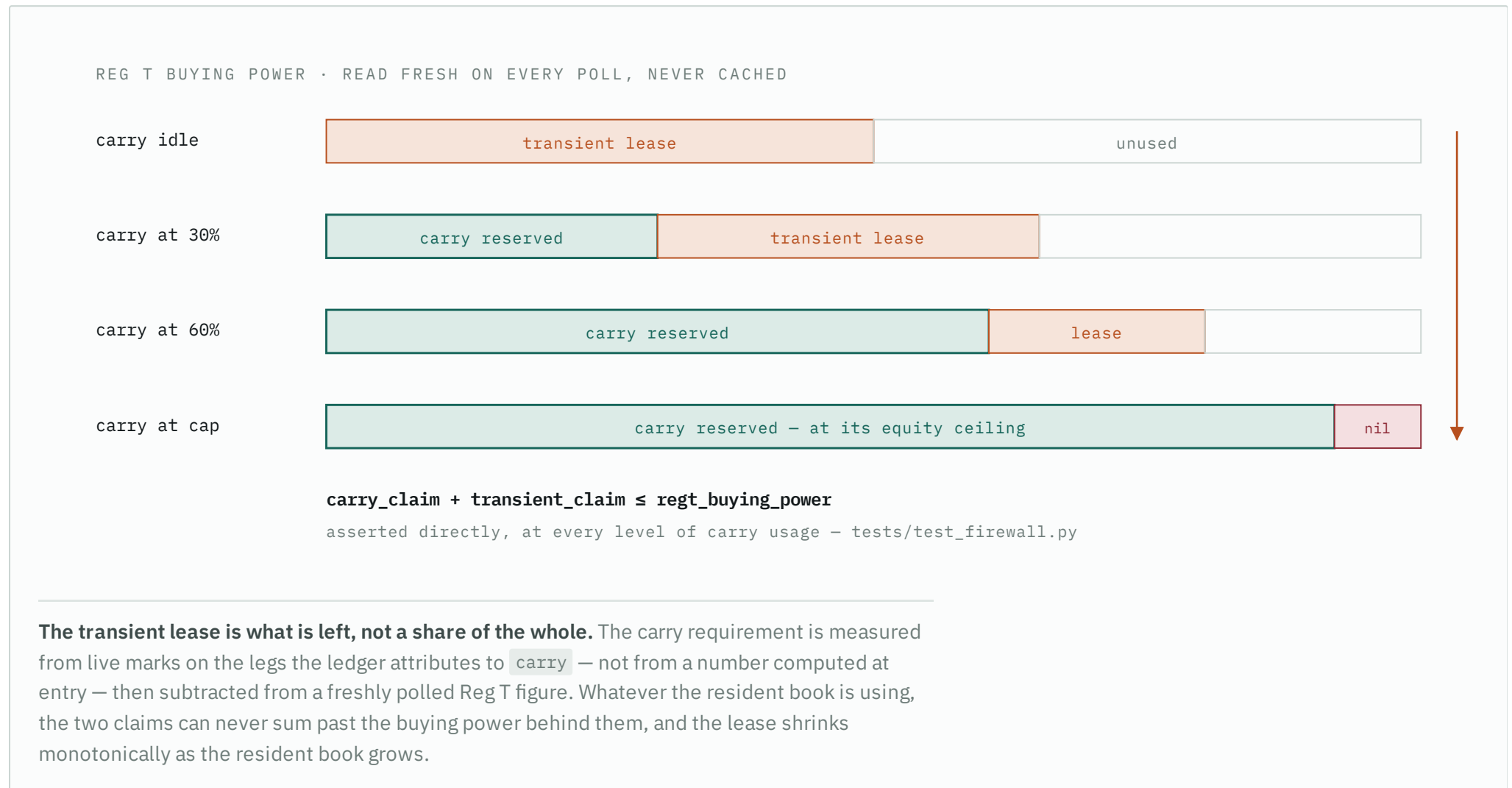
## THE CAPITAL FIREWALL

# The reservation comes off the top

Alpaca grants 4× day-trading buying power but only 2× Reg T overnight.

Intraday leverage still on the books at 16:00 is a broker-forced liquidation —

unrecoverable inside a one-week judged window. Two layers make the overlap impossible.



## The position ledger

Once the carry book became resident, “flat” stopped being a property of the account and became a property of a *book*. A persisted `symbol → book` map is what lets the cutoff tell them apart, and it survives a restart at 15:10 — otherwise the 15:15 liquidation would take the resident condor with it.

#### THE CONSERVATIVE DEFAULT

**Unattributed legs are treated as transient.** A leg the ledger has never seen is, by definition, not something the system deliberately chose to hold overnight. Closing it at 15:15 is the recoverable error; carrying it into the close is the one that isn’t.

#### What the cutoff actually guarantees

- **Liquidation is confirmed, not requested.** A 200 from `close_all_positions` means accepted, not filled — so it polls, four attempts, five seconds apart.
- **Working orders count as “not flat”.** A resting order that fills at 15:59 is unexpected exposure into the close.
- **Failure at 15:45 disables the transient books for the following session.** A book that needed rescuing does not get fresh leverage the next morning.

CARRY BOOK • SIGNAL STACK

## Four hard gates, not a score

A blended score would let a rich-IV reading paper over an earnings date. That is precisely the trade this strategy must never take, so every gate is hard and a

candidate that passes three of four does not trade.

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01 **Premium — is vol rich?**

IV rank says vol is high *for this name*. The IV–RV spread says you are being paid more than the stock has actually been moving. Both are required; either alone is a known false-positive generator.

```
iv_rank ≥ 0.70 AND (iv - rv) ≥ 0.03
```

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02 **Trend — is it going nowhere?**

Short premium is short movement. This is the *same* measurement that fires `momentum_debit_spread`, which is what makes the two strategies mutually exclusive by construction rather than by coincidence. A test asserts they never co-fire.

```
adx ≤ 25 AND |trend| ≤ 0.60
```

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03 **Event — hard exclusions**

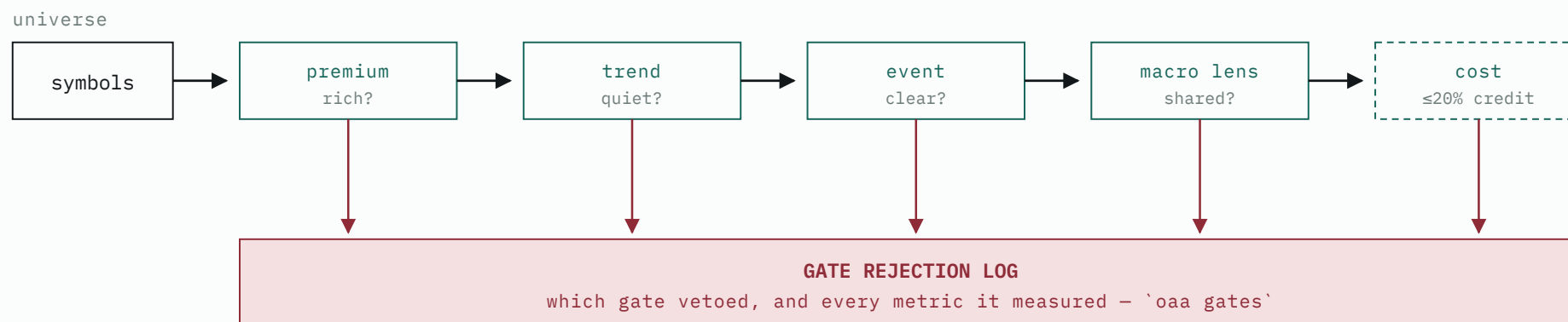
Earnings inside the expiry window, or an ex-dividend date with a short call. No model input, no scoring. IV is elevated *because* of the event, so that premium is fair — selling it is selling the one thing the market has priced correctly.

```
no earnings ≤ dte_max · no ex-div with a short call
```

## Macro lens — shared or idiosyncratic?

The question a z-score cannot answer, and the one place a language model genuinely earns its seat. Sector-wide IV elevation with no name-specific catalyst is **shared** — exactly the premium the thesis targets. A name repricing on its own news is **idiosyncratic**: the distribution has a fat tail the model cannot see. Veto.

guidance  $\neq$  stand\_down AND symbol not flagged



A refusal is an artefact, not a dead end: the log is the evidence that the agent reasons rather than fires.

**Every veto is written down.** The rejection log is the highest-value artefact for judging — it shows an agent *declining* trades and the exact measurement that stopped each one. A strategy that says “IV rank is 0.74 but earnings land in nine days, so no” reads far stronger than one that simply fires.



## Why 7–14 DTE, and not the conventional 20–45

Theta is steeply non-linear into expiry. At 30 DTE a five-session hold captures roughly 20–25% of the structure's remaining decay; at 10 DTE it captures the majority of it. **The judged window is the constraint, and the structure's life has to fit inside it.**

The cost is gamma, and it is a real cost — short-dated short-premium positions reprice violently when the underlying approaches a short strike. Three controls offset it partially, not completely: 10–16 $\Delta$  strike selection, the position-count cap, and the defensive exit on a short-strike touch.

### PUT THIS IN THE DECK

This is a deliberate trade of tail risk for realised P&L inside the window. It is the right call for a seven-day judged period and the wrong call for a live book. A judge who sees 10-DTE condors will either think you don't know the convention or that you knowingly departed from it — and the second impression is only available if you write it down.

## Why a 30% profit target, and not 50%

Fifty percent is standard practice for a book with unlimited time. This book has five sessions. A target that is rarely reached leaves positions open at judging, which means the P&L is an *unrealised mark-to-market on wide option spreads* — noisy, unflattering, and marked at a mid the account could not actually trade at.

Taking 30% converts decay into realised gains inside the window and produces more **closed** trades, which matters twice over: win rate, average hold and per-trade attribution

only exist for closed trades, and the equity curve becomes a curve rather than one flat line with a single mark at the end.

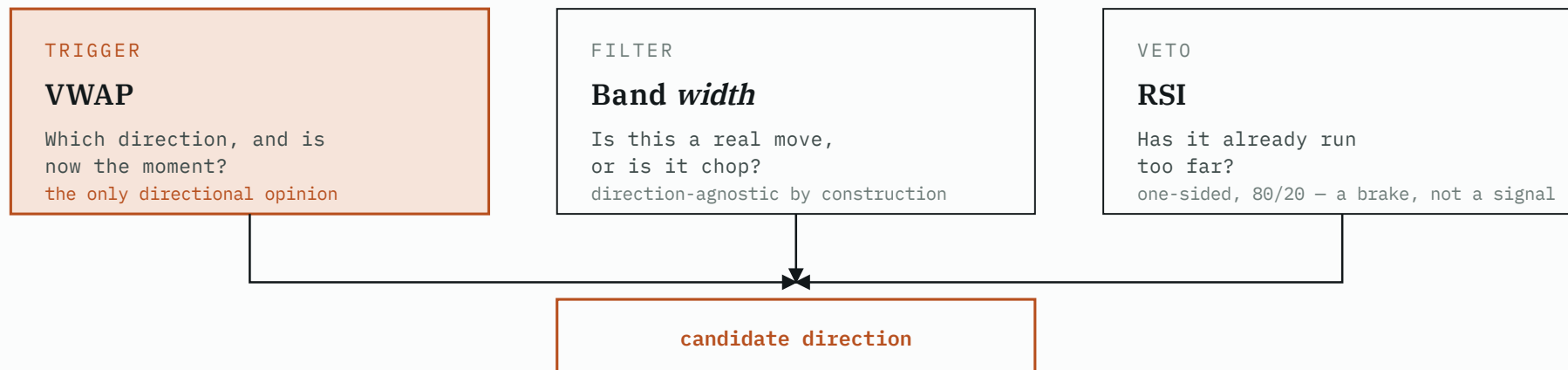
| EXIT TRIGGER                       | ACTION                                                    |
|------------------------------------|-----------------------------------------------------------|
| 30% of max profit                  | close                                                     |
| DTE floor (3)                      | close regardless of P&L — gamma rises sharply into expiry |
| underlying touches a short strike  | close the tested side                                     |
| loss reaches 2× credit             | close                                                     |
| macro lens flags the name mid-hold | close at the next open                                    |
| submission flatten                 | close everything, unconditionally                         |

INTRADAY BOOK • SIGNAL STACK

# Three indicators, three different questions

The critical design choice is that **only VWAP has an opinion on direction**. RSI and Bollinger *position* are mean-reversion measures and would contradict a VWAP momentum trigger on every single signal — price breaking above VWAP is simultaneously “momentum, buy” and “overbought, fade”.

A naive consensus gate across all three either never fires or fires incoherently. Give each a different job and they become orthogonal.



Because the three are non-overlapping, this fires at a usable rate rather than almost never – which is the failure mode of a three-way agreement gate.

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**Orthogonality is the whole trick.** Width, not position: band width is a volatility-regime measurement, so expanding bands mean volatility is genuinely picking up, and contracting bands during a VWAP cross mean chop. RSI is used one-sided at 80/20 rather than 70/30 – strong momentum runs carry elevated RSI for extended periods, and a tight threshold rejects exactly the trades this book exists to take.

### The constraint that shapes everything: spread cost

Target profit is 5–15% of premium. On a \$2.00 ATM option that is \$10–30 per contract.

| QUOTE WIDTH                | ROUND TRIP / CONTRACT | SHARE OF A \$20 TARGET  |
|----------------------------|-----------------------|-------------------------|
| \$0.01 SPY ATM             | \$2                   | 10%                     |
| \$0.02                     | \$4                   | 20%                     |
| \$0.05                     | \$10                  | 50%                     |
| \$0.10 typical single name | \$20                  | 100% — NO EDGE SURVIVES |

**Index products only.** Not a preference — arithmetic. This is not a universe-wide momentum system; it is a two-symbol system, because those are the only two symbols where the sums work.

### Catalyst confirmation — the non-generic layer

Everything above the catalyst gate ships with every charting platform on earth. A VWAP cross with no reason behind it is noise, and noise mean-reverts. A VWAP cross on a dated catalyst is a move with a *mechanism*, and mechanisms persist.

Three factors, blended after **rank-normalisation within each source** — raw units across sources are not comparable, and blending them unnormalised silently lets whichever source has the largest numbers dominate the weights.

| FACTOR               | WEIGHT | WHAT IT READS                                                                                                                                                    |
|----------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| News intensity       | 0.50   | Count and recency-decayed relevance in the window. Weighted highest because it is the <i>only</i> factor with a mechanism attached.                              |
| Breadth              | 0.35   | Share of the movers list going the same way. Load-bearing on an index universe: SPY rising on mixed breadth is one mega-cap dragging the tape, and that reverts. |
| Volume participation | 0.15   | Standing in most-actives, independent of the bar data.                                                                                                           |

Sources are Alpaca-native — `data news`, `screener movers`, `screener most-actives`. No scraper to maintain, no ToS exposure, and nothing to break on day three of a seven-day window. The path is deterministic by default: a model call inside a loop whose signal decays in minutes is a latency risk, so the LLM enriches this read rather than gating it.

#### POLARITY INVERTS

The macro lens's shared-versus-idiosyncratic judgement ports across from the carry book with the sign flipped. For a short-vol book a *shared* catalyst is safe to sell and *no* catalyst is ideal. For a long-vol directional book a shared catalyst is **tradable** — a broad move continues — and no catalyst is the **veto**.

### Option selection — where the agent adds value

Buying the ATM call is what a script does. Strike, expiry and *structure* conditioned on the expected move and the current surface is a real decision, and it is the most defensible

answer to “why does this need an agent at all”.

| EXPECTED<br>MOVE | IMPLIED<br>VOL         | SELECTION            | REASONING                                                     |
|------------------|------------------------|----------------------|---------------------------------------------------------------|
| small            | low                    | ATM, 0–1 DTE         | max gamma, cheap                                              |
| small            | elevated               | ATM, longer<br>dated | less premium burn if the move stalls                          |
| large            | low                    | slightly OTM         | better convexity per dollar                                   |
| large            | elevated               | debit vertical       | caps the cost of the elevated IV; sells the<br>expensive wing |
| any              | IV rank $\geq$<br>0.85 | NO TRADE             | paying for a move already priced in                           |

That last row matters: **the strategy declines to trade when the option is expensive relative to the move it expects.** That is a vol-aware directional strategy rather than a naive one, and it is worth a slide.

### Exits, and the asymmetry in them

Target 10% of premium, stop 15%, time stop at 20 minutes, close on a VWAP re-cross against the position, and unconditionally at the 15:15 cutoff. Note the stop is *wider* than the target — deliberately. Option premium is noisy and a tight stop gets hit by spread flicker alone.

The cost of that choice is a breakeven hit rate around 60%, which is a demanding bar. It is computed from the target/stop asymmetry and displayed against the actual rate, rather than assumed away.

#### THE ORDER PATH

## Only one component can say yes

The critic scores and writes reasoning. Partner adapters may veto. Neither can approve. Between the idea and the broker sits a deterministic engine, and execution refuses any ticket it has not stamped.

ADVISORY – MAY SCORE, MAY VETO, MAY NOT APPROVE

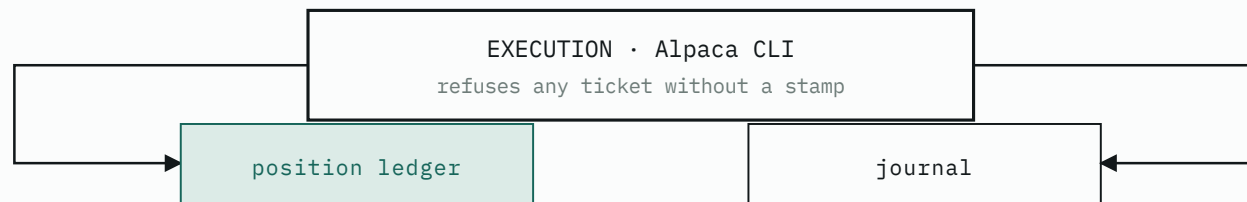


DETERMINISTIC – THE ONLY APPROVER

### RISK ENGINE

- 0 · firewall – wrong book, wrong minute → categorical refusal
- 1 · session state, halts, daily loss limit
- 2 · defined risk · computable max\_loss · 1-4 legs · no duplicates
- 3 · portfolio shape · 4 · sizing FROM MAX LOSS · 5 · aggregate
- 6 · cash buffer · 7 · time-of-day window

signed stamp



**Ledger registration is what makes 15:15 correct.** Every filled leg is attributed to its book at the moment it opens, which is the difference between a cutoff that liquidates the day book and one that liquidates the whole account. Where a venue cannot route a combo atomically the structure is legged *long legs first* and unwound in reverse on any failure – so no partial fill can leave an uncovered short.



INSTRUMENTATION

# Built from the first commit, not retrofitted for the deck

| ARTEFACT           | WHY IT EXISTS                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gate rejection log | Which gate vetoed each candidate and every metric it measured. The highest-value judging artefact: it shows an agent reasoning, not just an agent trading. |
| Modelled cost line | Gross P&L, modelled cost, net — side by side. Paper charges no fees and fills at mid, which flatters the intraday book severely.                           |
| Breakeven hit rate | Computed from the target/stop asymmetry and shown against the actual rate.                                                                                 |
| Equity curve       | Snapshotted every five minutes, so the curve is dense enough to be the deck’s primary visual evidence.                                                     |
| Macro lens archive | Regime, flagged symbols and themes, per cycle — the reasoning trail behind every stand-down.                                                               |

LIMITATIONS

# Four things that are true and should not be hidden

A judge who spots an unacknowledged paper-fill advantage forms a much worse impression than one who reads that the team modelled it.

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## **The momentum layer is conventional.**

VWAP and volume confirmation ship with every charting platform, and any edge in a bare VWAP cross was arbitrated long ago. The originality is the catalyst gate and the surface-aware option selection — not the trigger.

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## **Paper fills flatter the intraday book severely.**

Fills at mid, no queue position, no partials. Spread cost is the strategy's primary loss mechanism and it is exactly what paper does not simulate, so paper results there are close to meaningless as evidence of live viability.

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## **Five sessions is not a sample.**

With gates that reject most candidates and a six-trade daily cap, expect perhaps 10–20 intraday trades in total. That cannot distinguish edge from luck in either direction.

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## **Short-dated short premium is a deliberate tail-risk trade.**

7–14 DTE fits the judged window and would be the wrong call for a live book. The gamma is real and the offsetting controls are partial.

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Options Alpha Agents · branch **strategy-v2** · carry resident · intraday and opportunistic transient · entry cutoff 2 Sep 20:00 UTC · submission  
flatten 4 Sep 13:45 UTC